Will Doster

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EDUCATION

Massachusetts Institute of Technology **Cambridge, MA**
B.S. in Mechanical Engineering, Minor in Computer Science | **GPA**: 5.0/5.0 Class of 2027
Relevant Coursework: Design and Manufacturing, Thermal-Fluids Engineering, Mechanics and Materials, Dynamics and Control, Robotics, Machine Learning, Fundamentals of Programming, Algorithms, Numerical Computation.

PROFESSIONAL EXPERIENCE

SpaceX **Starbase, TX**
Tooling Intern June 2026 – August 2026
- Designed gas-spring-actuated flip decks, work access platforms with novel electrically actuated hinged staircases for over-the-barrel access, and a 7,300 lb-capacity spreader bar for Starship barrel tools using hand calculations and FEA.
- Owned projects from engineering drawings and supplier quoting through fabrication planning and build oversight.

Mecado **Cambridge, MA**
Machine Learning Intern January 2026 – February 2026
- Fine-tuned an LLM on semantic-to-CAD pairs, using supervised learning for CAD syntax and reinforcement learning for geometric reasoning. Programmed an NX plugin enabling end-to-end text-to-CAD generation in a live CAD environment.

Northrop Grumman **Allegany Ballistics Laboratory, WV**
Structural and Thermal Engineering Intern June 2025 – August 2025
- Performed advanced structural and thermal FEA with Abaqus on rocket motor components for a variety of load cases.
- Designed the first modular, 3D-printed ring igniter housing to reduce manufacturing costs and prevent nozzle clogging.

kW Mission Critical Engineering, Member of WSP **Atlanta, GA**
Mechanical Engineering Intern June 2024 – August 2024
- Designed large data center cooling systems by performing fluid and thermal load calculations, producing piping drawings.

PROJECTS

MIT AURA Lab: Undergraduate Researcher, SuperUROP **Cambridge, MA**
Flapping-Wing Robot Design August 2025 – Present
- Built the first robot designed to fly and swim with actively pitched wings, using a differential gearbox.
- Applying Bayesian optimization to search the flapping parameter space, improving force production and efficiency.

MIT Design and Manufacturing Robotics Competition **Cambridge, MA**
Competition Robot Design January 2025 – May 2025
- Designed, manufactured, and programmed a robot with an autonomous cascading arm and deployable lever flipper, placing 3rd of 150 competitors. One of two recipients of the International Design Competition award for outstanding originality.

MIT Liquid Propulsion and Controls Team **Cambridge, MA**
Nitrogen Back Pressurization System Design August 2025 – May 2026
- Led sizing of tanks and valve selection for a bipropellant rocket engine pressurization system using pressure drop analysis.

MIT Precision Machine Design Lab: Undergraduate Researcher **Cambridge, MA**
Flexure Design August 2024 – May 2025
- Designed and FEA-validated a variable-stiffness flexure via an adjustable boundary for artificial muscle characterization.
- Designed and manufactured a vacuum workholding system for a mill that met tolerances of ±0.001" on machined flexures.

Laser Ablation System Design/Controls November 2023 – May 2024
- Programmed laser motor controls using LabVIEW and C#, translating mirror angles into a prescribed 3D cutting path.
- Designed and manufactured a piston system to open the protective hood with a maximum force of 70 N.

LEADERSHIP AND SOCIETIES

Chi Phi Fraternity: *President (Previously: Vice President, Secretary)* May 2025 – December 2025
- Organized bi-weekly meetings for 40 members. Oversaw all house positions and handled conflict resolution.

MIT Men's Lacrosse Team: *Defense* August 2023 – Present
- Training 20+ hours weekly for competition on MIT's Division III team. 3x Academic All-Conference award recipient.

Pi Tau Sigma Mechanical Engineering Honor Society: *President* January 2026 – Present
- Serving as primary contact for national organization, managing chapter operations, member induction, and campus events.

Tau Beta Pi Engineering Honor Society: *Member* January 2026 – Present
Global Teaching Labs (Italy): *Teacher* January 2025 – February 2025
Boy Scouts: *Eagle Scout, Senior Patrol Leader* August 2015 – May 2023

SKILLS AND INTERESTS

Engineering: CAD, FEA, Drawings/GD&T, CNC Mill/Lathe, Metrology, Injection Molding, Welding, Rigging/Lifting
Software: Python, PyTorch, MATLAB/Simulink, Git, Arduino, C#/C++, Excel, LabVIEW, Office
Interests: Backpacking, Snowboarding, Music, Fitness, Food